



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: VI

Programme: B. Tech Information Technology

Course: Natural Language Processing

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33608	Course Outcome
CO1	Classify the ability to preprocess text data using NLP techniques.
CO2	Demonstrate syntactic parsing methods, including context-free grammars and dependency parsing
CO3	Analyze semantic structures and apply information retrieval techniques to extract meaningful insights from text.
CO4	Interpret language models and perform text classification using algorithms in NLP.
CO5	Evaluate real-world applications of Natural Language Processing

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define NLP and mention one challenge.	CO1	2	2
b.	What is POS tagging?	CO2	2	2
c.	What is POS tagging?	CO3	2	2
d.	What is bigram model?	CO4	3	2
e.	What is sentiment analysis?	CO5	2	2
Group B				

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain complete NLP pipeline with diagram.	CO1	2	5
b.	Compare stemming and lemmatization with examples.	CO1	2	5
c.	Explain BoW vs Word Embeddings (Word2Vec, GloVe).	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain CFG and parsing with suitable example.	CO2	2	5
b.	Describe dependency parsing algorithm.	CO2	3	5
c.	Explain Named Entity Recognition (NER) and its challenges.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Vector Space Model with example.	CO3	2	5
b.	Describe IR evaluation metrics (precision, recall, F1).	CO3	3	5
c.	Compare different approaches of WSD.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain N-gram models with probability examples.	CO4	2	5
b.	Discuss smoothing techniques (Add-One, Good-Turing).	CO4	3	5
c.	Explain Naive Bayes classifier for text classification.	CO4	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain extractive vs abstractive summarization.	CO5	2	5
b.	Describe Machine Translation approaches.	CO5	3	5
c.	Discuss ethical issues in NLP applications.	CO5	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create